



AIRBNB PRODUCTION-SCALE SYSTEM ARCHITECTURE

High-Availability Vacation-Rental Marketplace · Multi-Region Scalability & Low-Latency Blueprint

ACTIVE-ACTIVE MULTI-REGION

LAYER 1 · GLOBAL EDGE & CDN

Client Devices

Web (Next.js), iOS, Android

Cloudflare Anycast CDN & WAF

DDoS Shield, Bot Management, SSL Termination

Edge Compute & Caching

Edge SSR Hydration, Geo-Routing & Stale-While-Revalidate

Cloudflare Images / Dynamic Media CDN

On-the-fly AVIF/WebP Resizing, Responsive Thumbnails

LAYER 2 · API GATEWAY & MICROSERVICES

Federated API Gateway

- Envoy Gateway / Apollo Router
- Rate Limiting (Token Bucket)
- OAuth2 / JWT Session Auth
- OpenTelemetry Tracing
- Schema Stitching / Subgraphs

Listing & Content Service

Property details, rooms, photos, amenities & host attributes

Booking Engine (Core)

ACID transactions, Hold locks, Inventory isolation & checkout

Pricing & Promo Service

Seasonal pricing, coupons (10%), cleaning fees, multi-currency

Search & Discovery Service

Spatial geo-radius, date filters, amenity filtering & ranking AI

Reviews & Trust Service

4.95 badge calculation, guest reviews, fraud & spam audit

Payment & Payout Service

Stripe / Razorpay integration, Escrow hold, host payouts

Notification & Chat

WebSocket real-time chat, Email / SMS / Push triggers

Host & User Profile Service

Mirashya Homes profile, KYC, Wishlists, badges & Superhost

LAYER 3 · STORAGE, CACHING & EVENT STREAMING

DB Sharded Aurora PostgreSQL

- Multi-Region Active-Active Shards
- CockroachDB for Global ACID Bookings
- Read Replicas across 3 Continents
- Connection Pooling via PgBouncer
- Point-in-Time Recovery & WAL archiving

R Redis Enterprise Cluster

- Redlock for Distributed Calendar Holds
- Sub-millisecond Listing Cache
- Session & Rate Limit Counters
- Geospatial Indexes (GEOSEARCH)
- Eviction Policy: volatile-lru

S Elasticsearch + Vector DB

- Full-text & Geo-distance Queries
- Multi-attribute faceted filtering
- Pinecone/Milvus for AI Similarity
- Real-time Change Data Capture (CDC)
- Debezium sync pipeline from Postgres

K Kafka Bus & Object Storage

- Apache Kafka Event Streaming
- Booking events, price updates, analytics
- AWS S3 / Cloudflare R2 Raw Assets
- Automated Lambda Transcoding Pipeline
- Snowflake Data Lake for BI & ML models

LAYER 4 · MULTI-REGION INFRASTRUCTURE & DEPLOYMENT

AWS EKS Multi-Region Kubernetes

Auto-scaling Node Groups, Karpenter, Istio Service Mesh, mTLS inter-service encryption & zero-downtime rolling deploys

GitOps & Infrastructure as Code

GitHub Actions CI, ArgoCD Continuous Delivery, Terraform modular blueprints, Canary releases & automated rollback

Observability & Reliability Engineering

Datadog APM, Prometheus & Grafana, OpenTelemetry distributed traces, PagerDuty alerting, Chaos engineering (Chaos Mesh)